



Predictive Fault Detection in Robots Using Deep Learning

Project Proposal

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1. Initial Working

At the initial stage of this project, we started working on the basic components required for predictive fault detection. First, we explored different predictive maintenance datasets to understand the type of parameters that affect machine or robot failures. From this study, we identified important parameters such as temperature, Torque speed, acceleration, battery level, motor load, and task duration that can indicate the health of a robot.

After understanding the dataset, we began developing a **simple robot simulation in Python**. In this simulation, a virtual robot performs basic tasks while its internal parameters are recorded at each step. This data will later be used to train the deep learning model.

We also prepared the **initial data preprocessing code**, where the collected parameters are organized into a time-series format suitable for training models like **LSTM or GRU**. Basic experiments were performed to visualize the data and observe patterns related to possible faults.

2. Dataset Description

For this project, we are using the **Predictive Maintenance Dataset** available on Kaggle. The dataset contains **10,000 records** of machine operating conditions and failure information. Each record represents the state of a machine during operation and includes several parameters that can indicate the health of the system.

The dataset includes the following important attributes:

- **Air Temperature [K]** – Temperature of the surrounding environment.
- **Process Temperature [K]** – Temperature of the machine during operation.
- **Rotational Speed [rpm]** – Speed of the motor or rotating component.
- **Torque [Nm]** – Force applied by the motor.
- **Tool Wear [min]** – Duration for which the tool has been used.
- **Target** – Indicates whether a failure occurred (0 = No Failure, 1 = Failure).
- **Failure Type** – Specifies the type of failure such as tool wear failure, heat dissipation failure, or power failure.

This dataset is suitable for predictive maintenance because it provides **time-based machine parameters** and labeled failure **events**. These features will be used to train a deep learning model such as **LSTM** to learn patterns that appear before a machine failure occurs.

3. Development:

So far, we have completed the initial steps required for building the predictive fault detection system. First, we selected the **Predictive Maintenance Dataset** from Kaggle and studied its structure to understand the different parameters related to machine failures.

After selecting the dataset, we implemented the initial Python code to read and explore the data. The dataset was loaded using Python libraries such as **pandas** and **numpy**, and basic analysis was performed to view the dataset structure, number of records, and feature descriptions.

Next, we performed basic preprocessing of the dataset. Unnecessary columns were removed, check for **missing values** and handle them if present, and the important parameters like air temperature, process temperature, rotational speed, torque, and tool wear were selected as input features. The **Target** column was used as the failure label.

We also applied **data normalization techniques** to scale the feature values so that they can be effectively used for machine learning models. After preprocessing, the dataset was divided into **training and testing sets** for future model training.

Finally, we prepared the data in a format suitable for **time-series learning models such as LSTM**, which will be used in the next stage to predict possible machine or robot failures before they occur.

This work establishes the foundation for building the predictive deep learning model and integrating it with the robot simulation in the next phase of the project.

Libraries:

```
import os
import sys
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
import tensorflow as tf
--
```

Pre-Processing:

```
def load_data(path):
    df = pd.read_csv(path)
    return df

def inspect_data(df):
    print("Dataset shape:", df.shape)
    print("\nFirst 5 rows:")
    print(df.head())
    print("\nSummary stats:")
    print(df.describe(include='all'))

def handle_missing(df, strategy='median'):
    missing = df.isnull().sum()
    print("\nMissing values per column:\n", missing[missing > 0])
    if missing.sum() == 0:
        print("No missing values found.")
        return df

    imputer = SimpleImputer(strategy=strategy)
    numeric_cols = df.select_dtypes(include=[np.number]).columns
    df[numeric_cols] = imputer.fit_transform(df[numeric_cols])
    print("Missing values filled using strategy:", strategy)
    return df
```

Input features:

```
def select_features(df, features, label_name='Target'):
    cols = list(features)
    cols_available = [c for c in cols if c in df.columns]
    if len(cols_available) < len(cols):
        missing = set(cols) - set(cols_available)
        print(f"Warning: missing expected feature columns: {missing}")

    if label_name not in df.columns:
        alt = [c for c in df.columns if c.lower() == label_name.lower()]
        if alt:
            label_name = alt[0]
        else:
            raise KeyError(f"Label column '{label_name}' not found in dataset")

    X = df[cols_available].copy()
    y = df[label_name].copy()
    return X, y
```

Main:

```
def main(csv_path=os.path.join(os.path.dirname(__file__), 'predictive_maintenance.csv')):
    if not os.path.exists(csv_path):
        print(f"CSV file not found: {csv_path}")
        sys.exit(1)

    df = load_data(csv_path)
    inspect_data(df)

    df = handle_missing(df)

    features = [
        'Air temperature [K]',
        'Process temperature [K]',
        'Rotational speed [rpm]',
        'Torque [Nm]',
        'Tool wear [min]',
    ]
```

Output:

First 5 rows:

	UDI	Product ID	Type	...	Tool wear [min]	Target	Failure Type
0	1	M14860	M	...	0	0	No Failure
1	2	L47181	L	...	3	0	No Failure
2	3	L47182	L	...	5	0	No Failure
3	4	L47183	L	...	7	0	No Failure
4	5	L47184	L	...	9	0	No Failure

[5 rows x 10 columns]

Summary stats:

	UDI	Product ID	...	Target	Failure Type
count	10000.00000	10000	...	10000.000000	10000
unique	NaN	10000	...	NaN	6
top	NaN	M14860	...	NaN	No Failure
freq	NaN	1	...	NaN	9652
mean	5000.50000	NaN	...	0.033900	NaN
std	2886.89568	NaN	...	0.180981	NaN
min	1.00000	NaN	...	0.000000	NaN
25%	2500.75000	NaN	...	0.000000	NaN
50%	5000.50000	NaN	...	0.000000	NaN
75%	7500.25000	NaN	...	0.000000	NaN
max	10000.00000	NaN	...	1.000000	NaN

[11 rows x 10 columns]

Missing values per column:

| Series([], dtype: int64)
No missing values found.

References

- [1] <https://www.kaggle.com/datasets/shivamb/machine-predictive-maintenance-classification>
- [2] Forecasting failure rate of IoT devices: A deep learning way to predictive maintenance
<https://dl.acm.org/doi/10.1016/j.compeleceng.2023.108829>
- [3] Machine Learning Approaches for Predictive Maintenance in IoT Devices By Mazedur Rahman 1, *, Amir Razaq 1, Md. Tanvir Hossain 2 and Md Towfiq Uz Zaman
https://wjaets.com/sites/default/files/fulltext_pdf/WJAETS-2025-1388.pdf